



Automation in business research: systematic literature review

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Abstract

Automation has profoundly transformed the operational landscape of companies across various industries. As organizations strive to adapt to this rapidly evolving technology, it becomes crucial for practitioners worldwide to identify the most suitable automation tools and solutions for their unique business needs. A systematic literature review serves as a valuable tool to gain a deeper understanding of the historical context of automation and to explore previous findings in this field. This study aims to provide an extensive literary overview of the history of automation spanning the years from 1966 to 2021. In this research, a combination of bibliometric, conceptual, and theoretical network analysis methodologies are employed, with the aid of VOSviewer software, to analyze and visualize the patterns within the existing body of automation literature. By utilizing bibliometric analysis, this study will map the key scholarly contributions and identify the main research themes and concepts. The findings of this systematic literature review will provide insights into the historical progression of automation research and its interdisciplinary nature, highlighting the significant milestones, emerging trends, and knowledge gaps in the field. Building upon these findings, the study will propose a research agenda to advance the scholarly debate on automation.

Keywords Automation · Business research · Systematic literature review · Technology

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1 Introduction

The automation process has been developing for several decades now. The current technology has evolved rapidly thanks to artificial intelligence, which has enhanced process optimization, customer preferences, and boosted decision-making (Langer and Landers 2021). Although the first inventions in automation were not effective, one of the major breakthroughs occurred in the 1990s, when the business management process was fully integrated with the automation process. The issue stalled until the early 2000s when the invention of Apple's Siri revolutionized the robotic industry, transforming it into artificial intelligence software, and advanced the robotic process automation from physical robots. Since then, automation research has undergone rapid development, additional research attention, and high industry adoption.

Automation is a multi-faceted concept and toolbox, which involves the automation of end-to-end processes by combining several technologies (Davenport 2018), such as robotic process automation (RPA), digital forms, optical character recognition (OCR), intelligent character recognition (ICR), workflows, natural language processing (NLP) engines, and machine learning (ML), to achieve desired organizational goals. It seeks to offer an attractive complementary response, a new strategic capability that complements more traditional levers. It helps companies rapidly improve operational efficiency, achieving a significant return on investment while freeing up resources and capabilities to launch strategic programs (Bortolotti and Romano 2012).

2 Research purpose

Our systematic review aims to assess empirical findings on automation in business literature to date, present a summary of key research themes and methodologies, and highlight implications for further study and application. This work has two goals. First, we review the literature to understand the state of research today, specifically how the idea of automation has been investigated. Second, the research reveals gaps in automation literature for future research purposes. The knowledge gained from this study should make it easier to conduct additional business research in the field of automation.

3 Research questions

To the current knowledge of the authors, a systematic literature review on automation has not been conducted previously. This paper addresses the following research questions (RQs): RQ1. What constitutes the research developments in automation in business journals? RQ2. What are the existing research themes on

automation? RQ3. What are the gaps in prior business literature and the avenues of future research on automation?

4 Research implications

The first theoretical implication is that this systematic review of business studies related to automation enables the scientific business community to better understand the applicability and importance of automation. The study also allows business scholars to comprehend how automation literature has been shaped and is evolving. Second, our paper further concretizes the different findings and affordances offered by these studies. Furthermore, the business scholars attentive to automation could utilize our study to position their research by working on the potential gaps identified in this paper and develop research queries for future work. Research gaps refer to areas or topics within a particular field or subject that have not been adequately explored. It should also help scholars collaborate and publish their research in relevant journals. Third, the present study aids the business research community to comprehend the merits of the prominent theories in the business literature related to automation. This study is thus limited in scope to the business domain.

4.1 Study layout

The remainder of this study is organized as follows: section one is the literature review section that describes the progression in automation literature over the past 50 years; section two explains the research method employed to conduct this review; section three reveals the results; and section four presents the discussion and conclusion.

5 Literature review

Automation has been a consistent objective of industrial revolution from as early as the 18th Century. Starting from the onset of the Industrial Revolution, there have been deliberate efforts to enhance the ease and efficiency of doing work through automation of complex processes requiring heavy workforce involvement (Janssen et al. 2019). According to Janssen et al., industrial revolution has been driven mostly by need and the needs keep evolving over time. The continuous evolution has led to the most recent outcomes in terms of AI and Internet of Things, among others, which are characterized as serious in-roads into automation.

5.1 History of automation

The world has experienced four phases of the industrial revolution with the first having occurred in the late 18th and early nineteenth centuries. The process of manufacturing in this period changed to an optimized form of labor using steam,

steam-enabled engines, and various machine tools traditionally used by individuals from the manual labor system (Marengo 2019). In the early twentieth century, the second phase of the industrial revolution ensued with steel and electrical factories being introduced. The introduction of steel and electricity boosted the efficiency and mobility of factory machines for the manufacturers, thus improving productivity through novel concepts, such as assembly lines (Kumar et al. 2005). An important milestone for business process automation was 1913, when Henry Ford conceived of the moving assembly line, an invention that enhanced the production of Model T cars in mass (Kotorov 2003). The late 1950s witnessed the emergence of the third phase—computer technology integrated with electronics in factories—and digital technology and automation started gaining traction as compared to analogue and mechanical technology (Hitomi 1994). Companies could apply automation to manage contracts due to the process' consistency and error-free nature. Automation then became popular in the 1970s and 1980s, when most businesses started applying the process to automate their tasks.

5.2 Industry 4.0

Over the years, automation has made significant advancements with the aid of new technologies, including artificial intelligence (Makarius et al. 2020). Although mechanization and industrial robots have been commonly used in manufacturing industries for several decades, the emergence of AI has opened up new possibilities for automation (Davenport et al. 2020), particularly in non-manufacturing industries like healthcare, finance, and retail. With the help of machine learning and natural language processing, AI-powered automation is now capable of automating complex tasks such as data analysis, decision-making, and customer service that were once deemed too challenging (Frey and Osborne 2017). Automation has increased the efficiency of work and enhanced the quality of outcomes.

With the emergence of Industry 4.0, automation has become an even more important concept. Industry 4.0 technologies, such as the Internet of Things, big data analytics, and artificial intelligence, have enabled manufacturers to create highly flexible and adaptive production systems that can respond quickly to changes in the market (Ding et al. 2021). This level of automation has transformed the manufacturing landscape, enabling manufacturers to achieve unprecedented levels of efficiency, productivity, and profitability. Businesses could now even employ automation to their marketing, operations, analytics, sales, customer relationships, and planning. Business process automation lets businesses or managers execute strategies and employ tools automatically after commanding and organizing them (Herm et al. 2022).

AI-powered automation is among the trends widely researched under Industry 4.0. It has infiltrated all dimensions of business activity, including manufacturing, marketing, customer resource management, enterprise resource management, warehousing operations, and human resource management (Altaf 2022). In marketing, for instance, machine learning algorithms have been used to predict consumer behavior and to develop marketing messages and strategies aligned with the consumer behaviors (Altaf 2022). Similarly, in sectors such as healthcare, AI

automation has resulted in greater efficiency in service delivery due to the use of interactive technologies to foster an understanding of patient needs (West and Allen 2018; Benner 2017). Remarkably, AI-powered automation has been recognized as a massive transformational force in the world. There is increasing interest in research on AI automation due to the observed transformational effect of the technology and the expected changes in the business environment (West and Allen 2018). A lot of the completed studies on AI-powered automation focus on the relevance of it on different domains of corporate performance and the expected impacts of AI on the future of business in various sectors (Chao et al. 2018).

Few studies explore the ills that could be associated with AI. As a continuously evolving field, more research is still needed on AI-powered automation, especially on issues such as the psychological impacts of such automation on employees and the implications of increasing automation complexity—from basic to advanced to intelligent processes. The mentioned areas can be considered as gaps in literature that are deserving of greater research attention. In the domain of AI-powered automation, it is important to note that there is a noticeable lack of research addressing the ethical challenges associated with emerging tools and technologies. While the field has seen significant advancements in automation capabilities, there remains a gap in understanding and mitigating the ethical implications that arise from their use. This gap represents an area that requires further exploration and research attention. One such ethical challenge is the issue of privacy. As AI automation relies heavily on data collection and analysis, there is a need to investigate the privacy risks and develop robust frameworks to protect individuals' personal information. Research efforts should focus on exploring effective methods of data anonymization, consent management, and privacy-preserving algorithms to ensure that AI-powered automation respects individuals' privacy rights. Another significant area that lacks sufficient research is the examination of bias within AI automation tools. As these tools become more pervasive, there is a growing concern about the potential reinforcement of biases present in the training data. Addressing this challenge requires investigating bias detection and mitigation techniques to ensure fairness and avoid discrimination in automated decision-making processes. Moreover, the issue of accountability remains relatively unexplored in the context of AI-powered automation. As these systems become more autonomous and make critical decisions, there is a need to establish clear frameworks for assigning responsibility and accountability when errors or adverse outcomes occur. Research efforts should focus on developing mechanisms for transparency, interpretability, and explainability of AI automation processes to enable better understanding and oversight.

6 Research on automation from the 1970s to 2021

Over the past 50 years, there has been increasing interest in automation research. The topic of automation is categorized into substantial subsets of published works across multiple journals. According to Janssen et al. (2019), 4–11% of published articles in the last 5 decades have focused on automation. About 5–6% of all the published automation articles were authored within the last two decades, indicating that, as the

popularity of automation increases, so does the interest that researchers have on the subject (Janssen et al. 2019). In the International Journal of Human–Computer Studies alone, more than 200 articles have been published on automation and its implications on manufacturing processes and daily living (Janssen et al. 2019). The period between 1980 and 1989 had the highest number of articles published on automation, followed by 2000–2009 and then 2010–2019 (Janssen et al. 2019). Notably, these trends are evident since, in the late 20th Century, the increasing popularity of automation led to greater emphasis on research on the subject.

By adopting a systematic approach, our review seeks to provide a critical analysis of the previous studies, which, in turn, enables scholars to identify the future range of work in the automation field. Also, automation-related studies have been published in a wide variety of business journals with different aims, scopes, and target audiences. The progression of knowledge in the management-related field has led to an increase in interdisciplinary, fragmented, and interdependent research. Therefore, it becomes vital to conduct a systematic literature review instead of other types of narrative reviews that compel scholars to focus on specific topics.

7 Research method

To generate an up-to-date overview of existing business research on automation, and quantitatively assess the related literature, we conducted a systematic literature review consisting of the gathering, critical assessment, and synthesis of existing knowledge on a specific topic. The systematic literature review is an efficient method to identify research gaps (Paul and Criado 2020). The method is widely adopted in business research (Paul and Feliciano-Cestero 2021), as it offers many benefits, including the capacity to create adaptable databases of research works that are readily updated and analyzed (Pickering and Byrne 2014).

Our data was gathered from Scopus, one of the most comprehensive sources of academic publications (Mariani et al. 2022). It covers works published since 1966, indexing 12,850 journals in numerous fields. Scopus was chosen over Web of Science for two reasons. First, as researchers face a trade-off between data coverage and cleanliness, Scopus has been discovered to have a larger coverage (60% larger) than Web of Science (Zhao and Strotmann 2015). Second, bibliometric studies in business research often employ only one database to avoid data homogeneity problems that could arise when using numerous databases (Galati and Bigliardi 2019).

To search the database, we first identified three keywords related to our study: business automation, management automation, and automation. Subsequently, we ran a query using a combination of these keywords (adopting the Boolean operator “OR”) in the fields related to “title,” “abstract,” and “keywords.” We considered works published only in business journals until December 2021. While the last 10 years have witnessed rapid development in technology and automation, a systematic literature review with a broader timeframe ensures a holistic understanding of the subject matter. It allows for a comprehensive understanding of how automation has developed, identifying trends, milestones, and the impact of

technological innovations over time. This broader perspective helps establish a foundation for understanding the current state of automation.

Proceedings, book chapters, and books were dismissed from further consideration. To filter this data, we relied on a screening process. Documents were excluded on the basis of whether they are not published in English and/or not available for the project team.

Upon considering all hitherto-mentioned selection criteria, our search yielded 2007 research works. The data included authors' names, titles, countries of the corresponding authors, the total number of publications, citation counts (i.e., total citations, average article citations, and the number of citing articles with and without self-citations), journal sources, keywords, countries and regions, and author-level metrics. The workflow of the selection process (Moher et al. 2009) can be seen in Fig. 1.

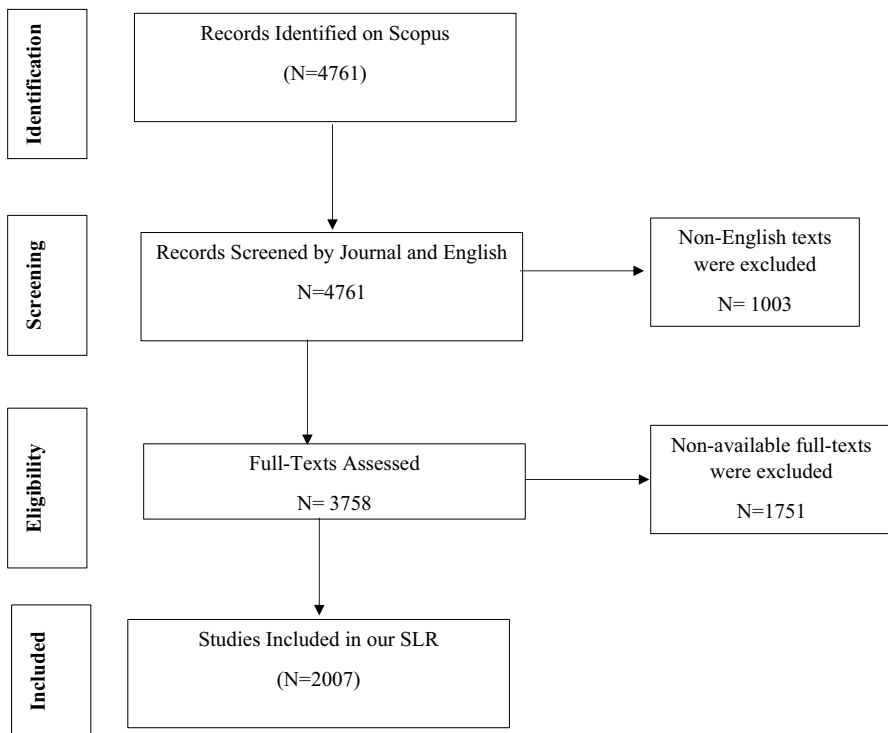


Fig. 1 Preferred reporting items for systematic reviews (PRISMA) statement. *Source:* (Self)

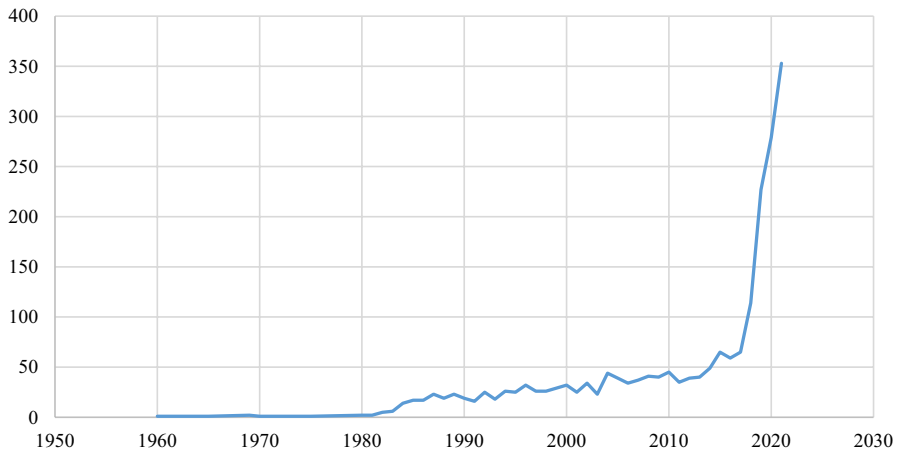


Fig. 2 The evolution of business research on automation

8 Findings

8.1 Descriptive analyses

In this section, we provide a descriptive analysis of the data gathered through our systematic literature review queries. Figure 2 presents the research evolution of business research on automation. In recent years, the total number of published articles has almost sextupled from solely 65 articles in 2016 to 353 in 2021, the year with the highest total number of published articles. This evolution suggests an exponential growth over the last years, which testifies to the growing scholarly interest in the topic. We note that most publications on the topic (66% of the articles) are accounted for from the period of 2011 to June 2021. In recent years, game-changing progress has been achieved in confronting several of the fundamental automation problems. Robotics and expert systems, telemetry and communications, and computer vision have all witnessed advancements. These innovations include the development and deployment of technologies for monitoring and controlling product and service production and delivery.

From Fig. 2 above, it is noted that the evolution of business research on automation was particularly rapid between 2010 and 2021. The number of publications grew almost linearly between 2017 and 2021, as observed from the graph, most likely due to increasing recognition of the impacts of automation and the emergence of AI-powered automation. It could be deduced that, from the 1960s to 2000, the focus of business research was solely on automation. In the latter years, however, research categorized under automation has largely been focused on AI and other Industry 4.0 evolutions and their impacts on the workplace. Additionally, the availability of technology could have increased the ease of undertaking research within the same period. Nonetheless, it is still difficult to predict the

primary purpose of research conducted through the years. Keyword analysis cannot analyze the article's content or interpret the primary purpose of each research.

Overall, the studies have been conducted in 61 countries. The greatest share of research has been carried out in the United States (34.6%). Business journals publishing on automation were also listed in our findings. They include a range of marketing journals (e.g. *Journal of the Academy of Marketing Science*, *Journal of Public Policy and Marketing*, *Psychology and Marketing*), business research journals (*Journal of Business Ethics*; *EuroMed Journal of Business*; *Business Horizons*), and management journals (e.g. *Academy of Strategic Management Journal*, *International Journal of Human Resource Management*, *European Management Review*).

8.2 Theoretical lenses

The abstracts were evaluated for the keywords “theory” and “model” to discern theories and models that the research on automation contributed to—in our systematic literature review—to explore the various theoretical lenses utilized to inform the research (Mariani et al. 2022). In total, 182 papers had an abstract that included a theory or model. There were 27 different theories initiated from those works, and Table 1 summarizes the most utilized ones, ordered by frequency of mention.

As shown in Table 1, the research topic of automation could be approached differently within various theoretical frameworks. Theories mobilized by business researchers also emerge from different fields. In each of these cases, various authors explained the theory they employed to support their interpretations and elucidations of the data. Next, we concisely explain the top 5 theories used in automation research.

- System Theory is the interdisciplinary study of systems. A system is a cohesive conglomeration of interrelated and interdependent parts that may be of natural or human origin (Rocchi 2000). Each system is bounded by space and time, influenced by its environment, defined by its structure and purpose, and expressed by its functioning (Rocchi 2000). A structure could be more than the sum of its parts if it expresses synergy or emergent behavior. General systems theory

Table 1 Prominent theoretical lenses on research related to automation

Theory	Sample article
System theory	Kazancoglu et al. (2021)
Game theory	Cao et al. (2015)
Decision theory	Heinrich and Klier (2015)
Computation theory	Simchi-Levi and Wu (2018)
Theory of planned behavior	Russel et al. (2020)
Institutional theory	Gozman and Currie (2014)
Stakeholder theory	Yang et al. (2017)
Information theory	Egorov et al. (202)
Constraint theory	Meneghetti et al. (2015)

consists in developing broadly applicable concepts and principles, as opposed to concepts and principles specific to a field of knowledge. It distinguishes dynamic or active systems from static or passive ones. Active systems are structures of activity or components interacting in behaviors and processes. Passive systems are structures and components that are being processed. For example, a program is passive when it is a disk file and active when it runs in memory (Rocchi 2000). The field is related to systems thinking, machine logic, and systems engineering.

- Game Theory is a formalized approach to the interaction between individuals. It models interdependencies to highlight the way in which strategic actions are constructed and respond to each other (Owen 2013). It is more beneficial to the decision-maker, as the reasoning it generates helps comprehend why and how the various parameters of a given context influence the strategies.
- Decision Theory is the result of several centuries of research on the formalization of chance and study of board games, on the analysis of economic and political problems, and managerial and psychological representations (Kaplan 1998). One of the main objectives of the development of this theory is to discover a reference framework for economic theories and management models of companies.
- Theory of Computation is a field of mathematical logic and theoretical computer science. Computation attempts to identify the class of functions computed with an algorithm, and to apply these concepts to fundamental questions in mathematics. An expert comprehension of what is computable—and what is not—permits one to grasp the limits of the problems that computers can solve (Sipser 1996).
- Theory of Planned Behavior postulates that, for human behavior to be effective, it must first be planned, hence the theory's name. Moreover, to be unambiguous, three types of factors are necessary (Ajzen 1991): (1) judgments about the desirability of the behavior and its consequences (attitudes toward the behavior), (2) considerations about the influence and opinion of others on the behavior (social norms), (3) beliefs about the individual's ability to perform the behavior.

8.3 Keyword co-occurrence and co-occurrence networks

We used a keyword co-occurrence analysis to find keyword co-occurrence networks in this study. The co-occurrence analysis approach described by Radhakrishnan et al. (2017) has been extensively used in literature reviews, especially in the technology sector. The most common areas of use of the technique are in scientometric and bibliometric research in diverse fields whereby mapping and clustering techniques find great relevance in following research patterns (Sedighi 2016). The approach helps in determining the links between different keywords that can be used in literature search. By evaluating the links between keywords, keyword co-occurrence analyses allow us to represent and understand the literature of a scientific topic (Mariani et al. 2022). Co-occurrence analysis identifies the connections that researchers make when selecting keywords for their papers in academic journals (Su and Lee 2010). Multiple tools exist for co-occurrence analysis, including Atlas.ti, T-LAB, TI.exe, and VOSviewer, all of which have different capabilities and different degrees of flexibility. In selecting the best tool

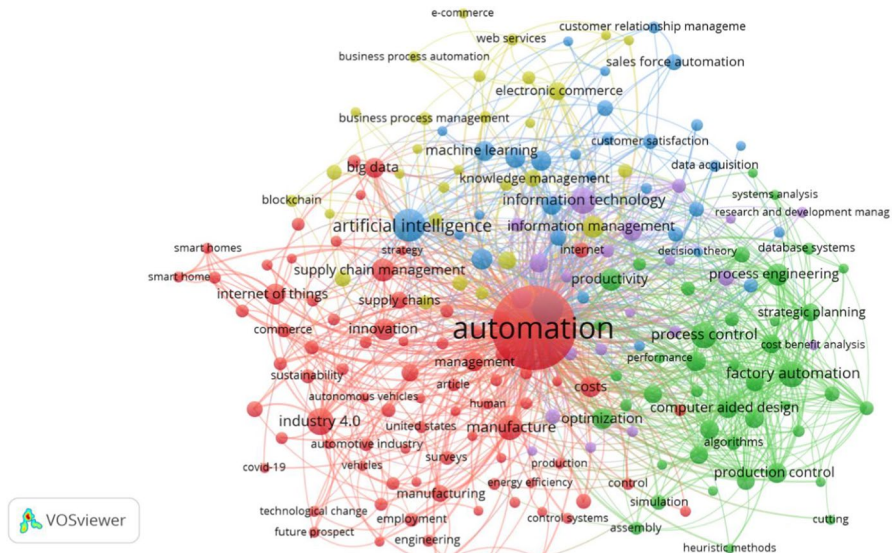


Fig. 3 Keyword co-occurrence networks in research on automation

for use, factors such as ease of use, diverse functionality, and multi-device usability are considered (Ferreira and Robertson 2020). The Van Eck and Waltman (2009) VOSviewer software was adopted to generate bibliometric analysis, as it has been extensively used in previous literatures with great success (Ferreira and Robertson 2020). VOS outperforms other co-occurrence analysis tools in ease of use, flexibility, and in multidimensional scaling in the creation of bibliometric analyses and maps (Van Eck and Waltman 2010). While this study did not include multidimensional scaling, VOSviewer was found to be satisfactory in terms of its capabilities.

Figure 3 depicts the outcomes of our keyword research. We selected the terms with the highest levels of occurrences as part of the keyword co-occurrence analysis, indicated by the size of the circles in the visual depiction (Mariani et al. 2022). We identified 5 topical clusters using VOSviewer clustering: (1) automation and production, (2) automation and technological innovation, (3) marketing and sales automation, (4) e-commerce and automation, (5) automation and information management. These clusters are also intelligible with the main uses and functions of automation within a company. This explains the emergence of these themes in business research. In fact, the integration of automation in production is crucial. Tasks such as packaging, distribution, or cutting are performed automatically to improve the company's profitability and productivity. Automation of production lines possesses many advantages in terms of competitiveness (Dash et al. 2019). Production costs and fast delivery are the most essential priorities for customers. Through automation, companies can achieve a significantly better price/performance ratio. Automation could also be integrated into marketing and sales. Marketing and sales departments use automation to automate online marketing campaigns and sales activities

to increase revenue and maximize efficiency (Kokemüller and Roßnagel 2012; Silva et al. 2021).

When automation is effectively used to handle repetitive tasks, employees are free to tackle higher-order problems, and human error is reduced. Furthermore, software and technologies permit e-commerce stores to computerize all their processes, designed to make life easier for e-commerce business owners and their employees (Cao et al. 2015). With workflow automation, e-commerce stores could computerize some of their most repetitive tasks within minutes of setup. This way, employees focus on more significant tasks and increase the productivity of the business. Finally, automation is strategic to manage a company's information assets, from emails to content shared with employees and customers (Cristani et al. 2018). Computerization captures, indexes, transfers, tracks, publishes, and archives inbound, outbound and internal correspondence. With automation, companies optimize their costs and work methods while preserving the integrity and security of information.

8.3.1 Automation and production

Production is one of the areas that has profited the most because of technological advancements in recent years. This is clearly because technology increases companies' production capacity, automates production processes, reduces costs, and/or plans production, assisting manufacturers to better manage their company operations in an utmost manner. The use of technology, such as hardware and software, to automate manufacturing procedures is identified as factory automation (Hoque 2000). There are different processes of production and, as a result, numerous types of industry automation exist to address various aspects. The first type of automation in production is hard automation; it is also known as fixed automation. This is frequently a system in which the manufacturing automation is established to produce a single product on a separate machine (Parthasarthy and Sethi 1992). Fixed automation, employed for large production and the possession of a high entrance barrier, is a type of automation. In hard automation, many sequences and industrial procedures are merged (Dunk 1992). As a result, altering the production style—once the machine is installed—becomes an exceptionally difficult task. This is especially true given the significant upfront cost of design and development. Assembly lines are an example of fixed automation, including material handling systems and chemical manufacturing processes.

A production system relying on programs to control is known as programmable automation (Huang et al. 2005). A program consists of a set of instructions that the computer reads, understands, and then executes. The ability to adjust preset automation is an important feature. Different product configurations or different orders are accommodated. Additionally, it introduces novel programs for the manufacture of other goods. For mass production, programmable automation is ideal (Singhal et al. 2007). Computer numerical control machine tools is an example of programmable automation including industrial robots and programmable logic controllers.

Another significant type of automation in production is the flexible automation. Programmable automation is comparable to flexible automation. It is meant to react swiftly to production changes, such as level of production. An employee presets a

computer system to operate the machine for it to run; a human–machine interaction is also feasible. The production and material handling operations of this factory are controlled by a central computer system (Schipper 2001). Subsequently, series production benefits from flexible automation. In addition, it is also a wonderful option if a variety of products are produced simultaneously. Assembly lines, robots, and material handling systems are examples of flexible automation.

Regarding its benefits to businesses, automation has evolved into more of a necessity than a luxury. One acquires a considerable competitive advantage by making use of the benefits it provides. Although automation may demand a considerable initial investment due to lower operational expenses, it offers tremendous long-term benefits (Udell et al. 2019). Depending on the job, one machine could assist (or replace) several people.

Since some industrial processes required individuals to toil in hazardous settings in the past, workplace safety has improved because of production automation (Kumar and Zhao 1999). Automation allows operators to be isolated from potentially hazardous procedures or materials.

The increased productivity not only allows the machine to maintain the same speed throughout the process but also permits it to function unattended all the time. As a result, the manufacturing process is capable of operating longer and more resourcefully (Brintrup et al. 2010). Consequently, one would be able to produce or focus more on new products uninterruptedly.

In general, product quality increases since machines are usually more precise than people for performing production tasks and maintaining an overall comparable level of product excellence (Castro-Lacouture and Skibniewski 2003). Reject rates are reduced when consistency and homogeneity are improved. Despite the advancements in manufacturing automation, there is still more to accomplish. As automation combines with robots, artificial intelligence, and machine learning, processes will become well-structured and/or organized.

8.3.2 Automation and technological innovation

Automation in businesses has evolved significantly through technological innovation between 1955 and 2021. Initially, automation relied on basic machinery, but the introduction of industrial robotics in the 1970s revolutionized manufacturing (Kong et al. 2017). Personal computers in the 1980s brought automation to offices, and AI and ML technologies enabled intelligent automation in the 2000s. The internet, cloud computing, and IoT advanced automation, leading to improved productivity and efficiency in businesses. From basic mechanization to intelligent automation powered by AI, ML, IoT, and cloud computing, businesses have witnessed significant improvements in productivity and customer experiences (Wang et al. 2020). The continuous advancements in technology drove automation further, reshaping industries and transforming the way businesses operate.

Automation is a set of technologies that permit the performance of numerous tasks while reducing human intervention. Automation can be used in all sectors and exists in various forms: (1) Robotic Process Automation (RPA), a specialized form of business process automation that replicates human computing actions, (2)

infrastructure automation, comprising of main tools like software that allows the creation of reproducible instructions, and processes in order to reduce human interaction with computer systems, (3) industrial automation, a form of automation that utilizes various equipment such as connectors (embedded, cable, etc.) to minimize human intervention in manufacturing processes, (4) business automation, aligning business process management and business rule management with modern application development.

Robotic process automation, or RPA, is an easy-to-use software technology that automates digital tasks. With RPA, it is possible to create software robots that could learn, reproduce, and execute rule-based business processes (Santos et al. 2020). RPA users could design robots by analyzing digital operations performed by humans, i.e. show them the tasks to be performed and, consequently, let them perform the work. Software robots interact with systems and applications of all types in a similar manner that people do, with a few differences, however: they are operational 24 h a day, work faster, and ensure complete accuracy (Santos et al. 2020). RPA is far more than just a technological advancement. It allows for the creation of a digital task force and, more notably, a command-and-control mechanism over it. RPA's objective goes beyond cost-cutting; it directly addresses a company's digital transformation by producing fresh value, enhancing service and product quality, and improving employee satisfaction by freeing them from repetitive and tiresome tasks (Dey and Das 2019).

Infrastructure automation is the use of software to create repeatable/similar instructions and processes with the aim of reducing human intervention and/or replacing it with computer systems (Byrd et al. 2008; Wesche and Sonderegger 2019). Automation software runs within the confines of these instructions, tools, or structures to perform tasks with minimal or no human intervention. Automation is a key component of IT optimization and digital transformation processes (Byrd et al. 2008). Dynamic, modern IT environments must be capable to evolve faster than ever before, and IT automation plays a critical role in this. In theory, any IT task could be automated to some extent. Subsequently, computerization could be integrated and applied to any task, from network automation to configuration management, and application deployment to provisioning of cloud, infrastructure, and standard operating environments (Johnson et al. 2021). The automation features and applications themselves could extend to specific technologies such as DevOps, and broader areas such as cloud, edge computing, security, testing, and monitoring.

The industries are being transformed by technology and automation, with robots and remote labor allowing manufacturers to improve productivity while decreasing the workforce (Millen and Sohal 1998; Emerald 2008). Furthermore, remote access solutions and cloud infrastructure have impacted management's ability to maintain employees productive from any location (Emerald 1993; Kushida et al. 2015). In analytics, artificial intelligence (AI) and machine learning systems are employed to construct predictive models that businesses may use to analyze performance, make accurate projections, and automate supply chains (Galende 2006; Engle and Barnes 2020). Artificial intelligence bots can now record, converse, and interact with consumers by forwarding requests to the relevant department (Nissen and Sengupta 2006; Saeed et al. 2020). Remote process automation can automate repetitive

operations by gathering information from bills, orders, or service requests (Järvinen and Taiminen 2016; Büchi et al. 2020). Data collection is also evolving rapidly. As a result, enterprises must supplement their cybersecurity capabilities using artificial intelligence to filter event notifications. The productivity gains acquired by technology and automation make it an appealing business strategy for any corporation aiming to develop a long-term, forward-thinking enterprise (Lichtenberg 1995).

Businesses have been pressed to employ technology and automation while investing in the latest innovations to stay operational and serve customers during this last year. Dealing pragmatically with the pandemic's urgency, organizations have swiftly adopted creative systems and solutions. Robotic process automation has become mainstream in commercial fields such as machine learning and artificial intelligence (Udell et al. 2019). Managers leveled up into embracing the quality and need for the value of process automation and complex information technology infrastructure by incorporating it into their businesses. The latest innovations in technology have continued to render some jobs obsolete. Technologies provide benefits such as enhanced production, faster manufacturing, better service, and more efficient material use (Peacock & Tanniru 2005; Radziwon et al. 2017). The latest technological innovations in business automation could alter any business or industry-predictive process, depending on the organization type (Kahle et al. 2020). For instance, it was more effective for companies to adopt it, especially if they had employed automation in their business activities fast enough to avert damages resulting from the pandemic effects.

Nonetheless, those who embraced the process late encountered difficulties in adapting and adjusting to the novel technology requirements for their business operations. The impact of technological innovations and automation on business industries' workforces cannot be overstated. For some employees, new prospects may compromise their job security; automation could replace semi-skilled humans in any context where predictable activities and procedures are structured (Thakkar et al. 2020). Manufacturing, hospitality, food service, and retail are all industries with a significant amount of scope for automation (Hutchison and Das 2007). Blue-collar and white-collar employment may be in jeopardy, inducing businesses to embrace skills through non-traditional training. Vocational schools, software training classes, technical certifications, on-site mentorship, and internships are all instances of vocational training. In other words, these employees are relentlessly aware of the most recent advancements in automation and technology, allowing them to support the company's operations creatively (Hernandez-de-Menendez et al. 2020).

8.3.3 Marketing and sales automation

This cluster emphasizes how sales force automation (SFA), customer relationship management (CRM), and marketing automation could be more accepted, acquired, and practiced by the companies' personnel. It mainly discusses how the sales force is motivated to use the sales force automation by citing the drive and readiness of the sales force as a major intention behind the success rate of SFA. Sales force automation refers to the employment of computer software, hardware, and telecommunication devices to enhance the use of information for improving the productivity of

the sales force and enriching customer relationships (Morgan and Inks 2001). Several articles in this cluster focused on the various factors leading to SFA adoption. The identity theory was used as a lens to competently comprehend the salespeople's perceptions leading to this technology rejection (Speier and Venkatesh 2002). Practices and outcomes of using SFA were also discussed.

CRM is crucial for the cultivation of better practices and customer relations in a sales force-customer context. CRM encompasses methodologies, technologies, and e-commerce capabilities used by companies to manage customer relationships (Landry 2005). Sales force support through sales business intelligence methods can boost the motivation of the salespeople to utilize CRM (Tseng and Won 2016). Cultural and social influence-related variables shall also be factored into the training and implementation of CRM (Baker 2014). Sales management technology, including sales force automation and customer relationship management (CRM) programs, has been increasingly utilized to manage knowledge and enhance productivity (Avlonitis and Pangopoulous 2010).

Marketing automation, whereby design meets technology to enhance the efficiency of marketing execution (Biegel 2009), was also deliberated as to how it should be purchased and implemented for achieving business transformation and business benefits. Marketing automation is an IT tool claimed to align marketing and sales system interfaces to improve lead qualification processes via 'lead scoring and nurturing' (Jarvinen and Taiminen 2016).

8.3.4 Automation and E-commerce

Automation has become increasingly important in the management of an e-commerce business. Studies in this cluster aimed at enhancing e-commerce, either by proposing automatic e-tendering systems that implement automatic negotiation processes over the Semantic Web (Du 2009), or through automating supplier selection procedures (Davidrajuh 2003). Some examined the impact of internet technology on sales organizations and identified forerunners for a prosperous hybrid sales arrangement (Thaichon et al. 2018). One of the main automation tools mentioned by the researchers in this cluster is business process management. The process management domain has been investigated in the e-commerce environment to provide different perceptions, methods, and software tools for designing and implementing more effective organizations based on processes (Margherita 2014). However, the complexity of business processes is seemingly huge. In consequence, various studies have ventured to solve their early re-engineering inefficiencies (in terms of cost and cycle times), thus redesigning the process altogether (Nissen 1998). Some studies tackled the management and control issues accompanying the reengineering of these business processes (Sia and Neo 1997), while others aimed at more efficient workflow processes, either through integration (Anyanwu et al. 2003) or through embedding formal analytics (Sun and Zhao 2013). More recent articles aimed at developing a system view of the business process management (Margherita, 2014; vom Brocke et al. 2016), providing a framework while building on existing business processes lifecycles (Macedo de Moraes et al. 2014), or proposing a model-based business process for better compliance checking (Becker et al. 2012, 2016). Chang

et al. (2019) suggested a blockchain-based framework and associated technology to reap the benefits of transparency, collaboration, and decentralization in the supply chain process design.

The literature identifies several roles that chatbots could assume in e-commerce. The diversity of roles is thus underlined (Pizzi et al. 2021), ranging from sales assistant to guide or conversation companion. Researchers mainly consider there exist three types of chatbots that correspond to three types of roles that a virtual agent could exert on a commercial website: recommendation agents, price search agents, and negotiation agents (Huang and Rust 2021). McGoldrick et al. (2008) identify three possible functions of online virtual sellers: the friend role (social role), the personal shopper role (recommender agent), and the helper role. Accordingly, several authors present the use of a chatbot on e-commerce as a strategy augmenting the satisfaction of the user and a means of attracting new customers by promoting positive word of mouth (Balakrishnan and Dwivedi 2021; Johnson et al. 2021).

8.3.5 Information management and automation

During the last decade, the usual management and operational practices have undergone profound alterations. Organizational modification has thus become a major concern for which many management methods have been proposed (Makarius et al. 2020). Companies must evolve to better satisfy their customers, improve their internal processes, and adapt the products and/or services offered to customer demand (Kunz et al. 2019). Their existence or endurance depends on innovation. Companies have experienced the effects of the integration of resourceful information and communication technologies in their business practices. In addition, the growing maturity of these technologies in the service of business practices has led to the evolution of business processes and logic (Batova et al. 2021). Nowadays, information management systems continue to support traditional requirements such as automation and coordination of the production chain and quality improvement of products and/or services offered. However, a new role has been assigned to them. It is the potential offered by information management systems to adopt a support role in the service of the company's strategy (Yin et al. 2020). Information, communication, and knowledge technologies (ICT) have thus been positioned as a strategic resource supporting organizational transformation (Kowalkowski et al. 2013).

In this evolving environment, organizations need to integrate new application solutions while simultaneously orchestrating the implementation of their activities and use of technological solutions in a global, integrated environment. Numerous articles have focused on understanding and exploiting the relationships between the systems to be built and business functions they will facilitate, guide, or even fully automate, either between the work processes implemented in an organization and/or its objectives. The first research works (Helo and Szekely 2005) are better known in the field of analysis and design of information systems, even if the notion of the business process has long been obscured by that of function. The second (Torres and Sidorova 2015), which seeks to specify the links between the company's objectives and the processes implemented, is more recent. In the analysis and understanding of an organization directed by the objectives, the engineering process must guarantee

the neutrality of the identified services concerning the organization's aims. Furthermore, in today's communicating and open forms of organization, which adopt particularly evolving management modes, the strategic planning of information management systems must lead to service-oriented architectures, capable of gradual evolvement in line with the strategy of the organizations they support. A strategic alignment is an approach with a goal to line up the information system strategy with the overall company's business plans (Ndede-Amadi 2004). This approach aims to reinforce the use-value of the information system and make it an asset for the company (Velcu 2010). The literature provides several frameworks offering a structuring in levels (or layers) of the organization (De Reuver et al. 2013). These frameworks enable the capacity to evaluate the organization and manage business processes and the information system at several levels. The top-level corresponds by consensus to a vision of organizational engineering, the intermediate levels to the implementation of application coordination, and the bottom level to the definition and implementation of new information management applications (Zur Muehlen 2004).

8.4 Research gaps

As our analysis indicated, automation is a recognizable field of study in business research. While the literature exhibits a growing level of research in the topic, some research areas are still untapped. First, the contexts under which these studies have been published are mainly in well-established and developed countries; less developed countries have witnessed no serious investigation of the subject. Consequently, future research should improve our comprehension of the adoption of automation in developing countries. Developing countries are characterized by certain determinants, including macroeconomic conditions and technological structures. Second, data security and privacy are probably crucial topics that have yet to receive the scrutiny they deserve regarding automation. Third, the consumer well-being literature has a lengthy tradition of management research, and it would be of interest to examine employees' well-being in an automation context—speculating, for instance, about the nature of the sales and marketing professionals' responses to marketing automation technologies. Fourth, there is a need to revisit research in relationship management between employees and executives in organizations with integrated AI used for automating the workflow. Finally, we list in Table 2 the main research gaps within every cluster.

9 Conclusion

The objective of this study was to review and systematize business articles on automation. It schematized existing studies and highlighted the importance of automation. The researchers' analysis confirms the theoretical validity of the concept. The current methodical literature review revealed a lack of real uniformity across papers concerning the theoretical models. This paper also underlines future research topics and their agenda, shaping the research areas that deserve attention in the business

Table 2 Suggested topics for future research

Cluster	Research gap
Automation and production	Exploring new models for production and distribution management in the context of automation Exploring automation in distribution management
Automation and technological innovation	Understanding cross-cultural comparisons of employees' experiences with technology Understanding age differences in perceptions and use of technology for work
Marketing and sales automation	Exploring the impact of automation on marketing teams' performance Exploring the impact of automation branding
Automation and E-commerce	Exploring the uses of automation in B2B e-commerce Explaining the effect of differences in personality, background, and experience to better understand the antecedents for human–chatbot relationships
Information management and automation	Exploring the challenges and opportunities in orchestrating resources for information management automation Exploring the effects of the newest trends in information management technologies on customer acquisition and retention

literature. To have a proper or appropriate number of publications in automation, and for the business research to progress further, this systematic literature review has assembled relevant publications, which ought to help future researchers publish additional impactful studies in this area.

We discovered that automation within companies at a practical level consists of transforming the internal organization of a structure, by entrusting a set of repetitive tasks, perceived as difficult and with low added value, to computer systems rather than to human resources. The objective of this automation is to improve productivity by reducing costs as well as all formalities, allowing the different teams to have additional time to focus on tasks with higher added value such as customer relations, analysis, and the follow-up of complex procedures. More precisely, automation permits enhanced effectiveness and efficiency on several business functions, such as sales management, marketing, information management, production management, and e-commerce management.

Some limitations should be acknowledged. First, the study merely examined and analyzed automation articles published in peer-reviewed academic journals. Consequently, books, theses, and conference proceedings were not included in the review. Future studies ought to consider this limitation, and might therefore wish to extend the scope of their studies. Second, our review comprised journals identified in Scopus; upcoming studies may contemplate examining other databases. Finally, in this study, we have considered only articles written in English. Automation has attracted global attention; therefore, studies conducted in other languages could have included auxiliary beneficial insights concerning this topic. Subsequently, forthcoming research may wish to concentrate on studies written in other languages.

Declarations

Conflict of interest The authors declared that they have no conflict of interest.

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